

PARTH ARORA

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Summary

Computer Engineering undergraduate (CGPA: 9.18) specializing in Backend Development, Machine Learning, and Generative AI. Experienced in building Retrieval-Augmented Generation (RAG) pipelines, multi-agent AI systems, and scalable REST APIs using Python, FastAPI, and LangChain. Solved 150+ Data Structures and Algorithms problems on LeetCode. Seeking a Software Engineering / AI-ML internship or full-time role to apply expertise in backend systems and applied machine learning.

Education

Thapar Institute of Engineering & Technology (TIET)

2024 – 2028

Bachelor of Engineering (B.E.) in Computer Engineering — CGPA: 9.18 / 10

Punjab, India

Technical Skills

Programming Languages: C++, Python, JavaScript

Backend Development: FastAPI, Node.js, Express.js, REST API Design, API Integration, Microservices

AI / Machine Learning: Machine Learning, Deep Learning, Natural Language Processing (NLP), LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Large Language Models (LLM), AI Agents, Prompt Engineering

Databases: SQL, MongoDB, Firebase, ChromaDB (Vector Database)

Core Computer Science: Data Structures & Algorithms, Object-Oriented Programming (OOP), Operating Systems, Database Management Systems (DBMS), System Design Fundamentals

Tools & Platforms: Git, GitHub, Docker, VS Code, Postman, Linux

Projects

EchoMind – Local AI Video & Meeting Assistant

GitHub Repository

Python, FastAPI, Whisper, LangChain, LangGraph, Llama 3, ChromaDB

- Built a local, privacy-first AI meeting assistant by designing a Retrieval-Augmented Generation (RAG) pipeline with Whisper, LangChain, and Llama 3, eliminating dependency on cloud APIs.
- Reduced audio transcription time by 96% (from 54 minutes to ~2 minutes for a 22-minute video) by integrating faster-whisper (CTranslate2 INT8 quantization) and optimizing audio preprocessing.
- Developed semantic transcript retrieval and conversational Q&A functionality using HuggingFace sentence embeddings and a ChromaDB vector database.
- Engineered a temporal retrieval layer enabling timestamp-based transcript queries, improving chronological search accuracy for long-form meetings.
- Exposed core functionality through a FastAPI backend, enabling modular integration with downstream applications.

NeuroLearn – Neuro-Adaptive Gamified Learning Platform

GitHub Repository

Python, FastAPI, LangChain, LSTM, React, Machine Learning

- Designed a multi-agent adaptive learning platform delivering personalized tutoring, multilingual instruction, and AI-generated educational content.
- Implemented LSTM-based knowledge tracing and spaced-repetition algorithms to drive adaptive review scheduling and difficulty personalization.
- Built ADHD-aware feedback logic by integrating eye-tracking attention data with a React-based mastery dashboard, improving learner engagement tracking.
- Developed scalable backend APIs using FastAPI to support modular learning workflows and personalized recommendation pipelines.

Problem Solving

Solved 150+ problems on LeetCode spanning Graphs, Dynamic Programming, Greedy Algorithms, Binary Search, and advanced Data Structures.

Relevant Coursework

Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Database Management Systems, Computer Networks

Achievements

3rd Place – MLSC Hackathon (NeuroLearn) | **Winner** – CTD Ideathon | **3rd Place** – OpenMic 3.0